From Lagging to Leading: The Measured Emergence of Standing Capacitance Behind Consumer Connections, 1997–2025

D. Hume

***Abstract*—Power systems are transitioning from net-absorbing (lagging) to net-generating (leading) reactive power at light load; the 2025 ENTSO-E Final Reports on the Iberian and North Macedonian blackouts identify excess capacitive reactive power among the causal factors. Using regulatory half-hourly metering at New Zealand grid exit points, we measure the transition end to end over 29 years. On a balanced panel of 123 distribution grid exit points (the same sites in every year), mean overnight reactive power fell from about $+466$ MVAr (lagging) in 1997 to about -451 MVAr (leading) in 2025 (about -31 MVAr/yr), crossing net-leading overnight in 2016—measured, not a metering artefact, with its intensity and reach unchanged when every contaminated grid exit point is removed. A first-principles charging calculation with no fitted cable coefficient puts the demand-side share of the drift at about 80% (physical band 75–88%) over 2013–2025; two statistical methods corroborate the direction. The deepening is as large at the flat evening peak as overnight—the signature of standing shunt capacitance, not changed operating draw—and it is proportional to connections served, with nothing left at zero connections: about 300 VAr per connection added over 2013–2025, crossing zero in 2012, at a rate still accelerating. The connected load fleet is adding standing distributed capacitance—consistent with the mandated electromagnetic-compatibility filter capacitance in every switched-mode device (a labelled hypothesis); both of the test's inputs are records any operator holds. The reversal relocates the binding voltage-security question to the overvoltage direction, where the defences, margins, and load models of practice are thinner or act the wrong way. We set out what this implies for the load models, security margins, and emergency schemes of power-system practice.**

***Index Terms*—Reactive power, power factor, overvoltage, leading power factor, shunt capacitance, distribution cabling, measurement, grid exit point, load composition, load modelling, power-system analysis.**

I. Introduction

POWER systems internationally are shifting from net-absorbing (lagging) toward net-generating (leading) reactive power at light load, as the connected load fleet modernises toward power-electronic devices and networks add underground cabling. The direction of the transition is not in dispute, and this paper does not claim to be the first to observe it. What has been missing is, first, a long *measured* record of the transition observed end to end at supply-point resolution, and second, a measured separation of how much of the drift is the load fleet and how much is the network. This paper supplies both for a single national system, connects the result to a failure mode that has recently moved from theory to realised blackouts, and closes with what the reversal implies for the analysis and defences that were built for the opposite regime. The trail ends somewhere unexpected: at the electromagnetic-compatibility filter inside every mains-connected device—capacitance that one part of the regulatory apparatus requires and no other part counts—advanced here as a labelled hypothesis with its decisive test identified.

A. A global reactive transition, already observed

The phenomenon is well attested across jurisdictions. In Great Britain, Kaloudas *et al.* [1], [2] documented a steep decline in reactive demand at minimum load, with distribution networks beginning to export reactive power to the transmission system; in France, the transmission operator attributes off-peak overvoltage episodes to a three-factor mechanism of distributed generation, underground cabling, and evolving final consumption [3]; in Australia, Powerlink Queensland links a rising likelihood of overvoltage events partly to more energy-efficient appliances alongside rooftop solar [4]. At device level, Hannagan *et al.* [5] found the majority of modern household appliances present a leading power factor, and international expert reviews reach consistent conclusions about modern lighting and power electronics and about the error introduced by constant-power-factor load modelling [6], [7]. No consumer-device standard requires a unity fundamental-frequency (displacement) power factor: IEC 61000-3-2 limits harmonic current only [8], and the European Demand Connection Code remains the only international instrument addressing reactive exchange at the demand connection point [9]. Rising capacitive export to the extra-high-voltage network has likewise been reported in continental Europe [10].

B. Prior measurement and decomposition

The closest measured system-level series is the British one [1], [2]: roughly eight years (about 2005–2013) of reactive demand at grid supply points, projected forward from load-composition modelling and scenario projection—neither a multi-decade measured record nor a quantified demand-versus-network share of the realised drift. Boundary studies in the Netherlands [11], Denmark [12], and Slovakia [10]

D. Hume is with the Electricity Authority, Wellington, New Zealand (e-mail: corresponding author details on submission). ORCID iD: https://orcid.org/0009-0006-6920-0598.

This is descriptive analysis of public regulatory metering and information-disclosure data. The views expressed are the author's. No external funding was received. The analysis informs, but does not advocate, a live reactive-power pricing policy question (Section VI-D).

A reproducibility package (the analysis notebooks that compute every number and figure in this paper from public data) accompanies the work.

characterise or simulate capacitive export at the transmission–distribution interface; none is a long measured record and none attributes a quantified share.

The single study closest to the present decomposition is Bosisio *et al.* [13] on the Milan distribution network: measured 15-minute boundary data at 37 high-voltage/medium-voltage transformers, naming the same three-term balance used here (customer reactive power, cable inductance, cable capacitance)—in the indexed English-language literature, to our knowledge, the only prior study to articulate this split explicitly on measured boundary data. It differs on three points that matter: two years (2019 against 2020, across a COVID-19 demand shock) rather than a multi-decade drift; no numerical share for the three terms; and a cable-dominant attribution, with reduced demand amplifying rather than causing it—on which term dominates, its reading runs opposite to ours. We treat it as complementary: it confirms the framework on measured data for one heavily cabled city, its cable-dominant reading is consistent with our cohort result that cable intensity shapes where net-leading territory is reached first (Section IV-A), and it leaves open the national, multi-decade, quantified attribution.

C. The realised risk

In 2025 two power systems suffered major blackouts whose official investigations found a leading-reactive/overvoltage mechanism at their core: the ENTSO-E Final Reports (both published 2026) on the Iberian Peninsula event of 28 April 2025 [14] and on the North Macedonia event of 18 May 2025 [15], detailed in Section VI-C. What matters here is what those investigations could not do. Both reports attribute the overnight capacitive surplus to network-side sources and treat low demand as a magnitude rather than as a fleet whose reactive character has itself changed; neither distinguishes or quantifies a demand-side composition shift from network charging. These are not criticisms of investigations that reasoned from the data available to them; they identify the space a long measured record can fill.

D. Contribution

This paper makes a layered contribution.

Contribution 1 (measured trajectory). Using regulatory half-hourly active and reactive power at New Zealand grid exit points, we measure the system's reactive character at supply-point resolution over 29 years (1997–2025) on a balanced panel—the same grid exit points in every year, so sites entering or leaving the record cannot manufacture the trend—characterise its rate and crossing, and show the trajectory is a measured phenomenon and not a metering artefact. The computation reuses settlement metering that any operator holds, so the same measurement is available to any system.

Contribution 2 (decomposition of the drift). We decompose the drift into a demand-side load-composition component and a network cable-charging component. The magnitude is carried by a first-principles charging calculation that computes each network's cable charging directly from its disclosed circuit kilometres (Section II-D) and so fits no cable coefficient; evaluated over the 2013–2025 disclosure window, it puts the demand-side (*organic*) share of the drift at about 80% (physical band 75–88%), with two statistical methods corroborating the direction over the full 29-year record but not themselves pinning the share. We report a measured direction and dominance and do not claim a point-identified split.

Contribution 3 (the term identified). We then identify what the demand-side driver is physically doing. The deepening is load-independent—as large at the flat evening peak as overnight—so it behaves as standing shunt capacitance, permanently energised, not as a change in how the load draws reactive power while operating. And it lives behind the connections: across grid exit points the 2013–2025 deepening is proportional to the number of connections served, with nothing left at zero connections; refitted year by year, each connection's standing contribution sits slightly on the inductive side of zero at the start of the record, crosses zero in 2012, and grows every year thereafter—about 300 VAr per connection added over 2013–2025, agreeing across two independent routes—and the accumulation rate is still rising (Section V). The load fleet is accumulating standing distributed capacitance, not merely drawing fewer lagging vars. We advance the mandated electromagnetic-compatibility filter capacitance as its candidate device layer (a labelled hypothesis), identify the decisive low-voltage measurement that would settle it, and note that both of the test's inputs—settlement metering and connection counts—are records any operator holds.

The significance is what the reversal does to analysis: the direction of the binding voltage-security question reverses; the automatic last-resort defences and the margin tooling of the undervoltage era do not transfer; and the constant-lagging-power-factor load archetypes of planning and protection studies no longer describe the light-load fleet (Section VII)—a qualitative synthesis, not a defence-scheme or load-model redesign, which remain outside the paper's scope. Section III measures the trajectory; Section IV attributes it between fleet and network; Section V identifies the term and tests its scaling; Sections VI and VII set out the overvoltage consequences and the analysis implications; Section VIII collects future work.

This work extends the author's EEA 2025 conference paper [16], which described the 2009–2024 aggregate trend qualitatively. The present paper adds twelve further years of baseline back to 1997, in the firmly lagging regime; grid-exit-point spatial resolution; a quantified rate and crossing with an explicit artefact test; the decomposition of the drift; the connection-scaling identification; and the two post-EEA ENTSO-E Final Reports as an independent evidentiary anchor. Consistent with its descriptive scope, the paper reports the measured trajectory and the decomposition and notes that they inform a live reactive-power pricing question, without making any policy recommendation.

II. Data and Method

A. Sign convention and primary variable

Reactive power carries a direction that is the entire point of this study. We use the convention $Q < 0$ for *leading* (capacitive, voltage-raising) and $Q > 0$ for *lagging* (inductive, voltage-lowering). We lead throughout with the signed reactive power in MVAr, which is additive across supply points and has no discontinuity at unity, and present power factor only as a familiar comparator. Power factor is non-additive, so where a system power factor is quoted we sum P and Q across the panel first and derive $\mathrm{PF} = \Sigma P/\sqrt{(\Sigma P)^2 + (\Sigma Q)^2}$ rather than averaging power factors.

B. The grid-metering record and the balanced panel

The primary data is half-hourly real and reactive power at every grid exit point (GXP), the substations where the national transmission grid hands electricity to the local distribution networks, from 1997 to 2025, published by the New Zealand Electricity Authority on its Electricity Market Information platform. Each per-year file is half-hourly for the whole year, indexed by trading date and trading period (48 half-hour periods per day), with a real-power and a reactive-power column for each GXP. Over the record, 238 distinct GXPs appear; a balanced core of 132 is present in all 29 years.

We reduce each GXP's year to a small set of features, of which the headline quantities are the average overnight and evening-peak reactive and real power. Following the prior EEA analysis, the overnight window is trading periods 6–10 (approximately 02:30–05:00, the deepest part of the night and the lowest demand) and the evening peak is trading periods 36–38 (approximately 17:30–19:00). We also record, for each GXP-year, the fraction of overnight half-hours that were leading. The full feature table is 5,261 GXP-year rows, one per GXP per year.

The single largest way a 29-year trend could be spurious is a change in the set of measured GXPs over time: new urban, cable-heavy sites appearing and old rural ones dropping out would manufacture a leading trend from bookkeeping alone. The primary defence is the *balanced panel*: only the 132 GXPs present in every one of the 29 years, so any change is a change in the same sites rather than a change in which sites are measured. We call that turnover—grid exit points entering and leaving the measured set—*churn*; the balanced panel is churn-controlled by construction ("balanced" in the panel-data sense, the same sites in every year, not the three-phase sense). We follow one rule throughout: *trends on the balanced panel, snapshots on the full panel*. The paper's subject is the distribution networks; restricting the balanced panel to demand networks (excluding a handful of generator, industrial, and rail connection points) leaves 123 GXPs. We report both the 132-point and 123-point figures and keep them labelled; they are the same trend at the same rate (Section III). The residual churn bias is signed: the demand grid exit points excluded from the balanced panel (post-1997 entrants and exits) are by 2025 on average *more* leading than the panel (median -0.28 against -0.19 MVAr/MW) and drift faster (-0.020 against -0.018 MVAr/MW/yr), so the balanced panel, if anything, understates the national drift.

C. Metering provenance and accuracy

The active and reactive power at each grid exit point is settlement-grade revenue metering owned by the national grid owner and system operator (Transpower), the registered metering equipment provider for these points. Grid metering is the highest-accuracy category of the regulatory Metering Code [17], certified by an approved test house and periodically re-certified and inspected. The present fleet is Schneider PowerLogic ION8800A meters, Class 0.2S for active energy under IEC 62053-22 [18], four-quadrant (import and export active and reactive), with Class A power-quality measurement under IEC 61000-4-30 [19], installed across roughly 150 grid sites in 2010–2013, replacing legacy PSI Quad4 meters dating from about 1994 [20]; the reactive series thus rests on four-quadrant revenue metering for the modern record and on the older, coarser Quad4 instrumentation for the early record.

At a grid exit point the active-energy channel is held to Class 0.2 while the reactive channel is held to the looser Class 2–3 (about ± 2–3.5%) of IEC 62053-23 [21], with reactive accuracy specified only above 20% of rated current; as a share of the reading, reactive measurement is least accurate at light load and near-unity power factor, where the reactive component is small while current-transformer phase-angle error is largest. This asymmetry is the international norm, because active energy is the settled quantity. The reactive energy of the metering standards is the fundamental-frequency (*displacement*) quantity of IEEE Std 1459 [22]: the reactive power set by the phase angle between the voltage and the 50-Hz component of the current. The harmonic currents drawn by power-electronic loads add apparent power (distortion power) but accumulate no reactive energy on the meter, so the trend measured here is in fundamental-frequency reactive power—the component that drives the voltage rise of Section VI—with harmonic distortion out of scope.

Neither the accuracy classes nor the one metering transition in the record can account for the measured drift; the full bounds, timing arguments, and test procedures are carried in the extended version [23], and the results are these. Random error is bidirectional, averages down under aggregation, and cannot manufacture a near-monotonic 900-MVAr swing through zero; below 20% of rated current the reactive class specifies no bound at all, which is why the one error that does *not* average down carries an explicit bound rather than a class limit. That error—a measurement-chain phase displacement of the form $P \sin \beta$, concentrated in exactly the light-load window this paper measures—is bounded, even for a coherent half-degree displacement across the whole panel, to a near-static offset under 20 MVAr, some fifty times smaller than the swing, moving the crossing year by about half a year at most and contributing about 0.1 MVAr/yr of apparent drift. The staggered 2010–2013 replacement of the legacy fleet was an *upgrade* to four-quadrant metering, so its possible bias runs in the conservative direction: under-recorded leading reactive power on the older fleet would delay, not advance, the measured crossing; leading flows were about 1% of overnight half-hours at the 1997 anchor; the crossing itself post-dates the rollout, so the leading regime is observed wholly on the modern fleet; and the per-MW drift within the modern-fleet era alone (2014–2025) is -0.019 MVAr/MW/yr against -0.018 over the full record, if anything steeper. A test for a synchronous system-wide re-base finds none (Section III-

B). We therefore treat the multi-decade *drift* as robust while regarding the standing reactive *level* on any single night as carrying a bounded measurement uncertainty—tens of MVAr at most, a few per cent of the 2025 total—distinct from the composition question of Section IV.

D. The asset (parameter) record

To describe each network's physical make-up we assemble a per-company, per-year table from the Commerce Commission's regulated information disclosures: circuit length split into overhead and underground kilometres at each operating voltage (Schedule 9c, disclosed consistently from 2013), capacitor-bank counts (Schedule 9a), demand and connection counts and embedded generation (Schedule 9e), and the regulated asset base (Schedule 4). The disclosures are mapped to the EMI network codes through a 29-year validated correspondence, time-aware because ownership itself changed under the record: 34 demand grid exit points (32 of them in the balanced panel) changed network code, almost all from the breakup of UnitedNetworks and the onward sale of its Wellington network (successor codes appear 2003–2013), plus one Nelson-region boundary change. The metering panel is attributed per year from the metering files themselves, so the trajectory follows every change; the cable-dose analyses of Section IV are run under both a per-year and a physical-continuity (successor-network) attribution, with the sensitivity reported there. Two data-quality choices are made explicitly: we retain only the "All"-network roll-up rows (multi-network companies are not double counted) and we flag a 2014–2015 definitional break in the capacitor counts (a widening of reporting scope, not new plant). The capacitor series is counts only, from 2013; we use it as a direction-only proxy—a stand-in that can indicate direction, not size.

E. Confounds in the metered reactive power

A grid exit point meter records the *net* reactive power at the boundary, which is not in general a clean distribution offtake:

$$Q_{\text{metered}} = \underbrace{Q_{\text{load}} + Q_{\text{cable}}}_{\text{the subject of this paper}} - Q_{\text{embedded gen}} \pm Q_{\text{TP plant}},$$

where the last two terms are *confounds*: signals mixed into the same metered reading that are not the subject of this paper. An embedded *synchronous* generator (hydro, geothermal, or cogeneration) regulates voltage and sets its reactive output by the voltage it sees rather than by local load; and at some substations a Transpower voltage-regulating device (a static var compensator or STATCOM, a synchronous condenser, or a switched capacitor or reactor bank) sits on the same metered bus as the offtake. One behavioural diagnostic catches all of these, whichever party owns the plant—a network's own large regulating compensation would be flagged identically: a clean distribution offtake's reactive power tracks its load, so a straight-line fit predicting Q from P explains an appreciable share of its variation ($R^2 \approx$ 0.3–0.5), whereas voltage-regulated reactive is decoupled from load ($R^2 \to 0$, with a large residual the load does not explain). Computed on three spread-out years (2016, 2019, 2022) so that a flag reflects a stable trait, the screen isolates six decoupled grid exit points: ISL0661 and BRY0661 (Islington and Bromley, Christchurch, where Transpower voltage-regulating plant—static var compensators, switched capacitor banks, and shunt reactors—shares the metered buses), PEN1101 and HOB1101 (Auckland), BPE0331 (Bunnythorpe), and TWH0331 (Tuai, with the Waikaremoana hydro stations); the site-level screen is reproduced in the accompanying notebooks [23].

This matters overnight specifically: restricting to the overnight minimum removes the *solar* confound—photovoltaics are dark at 03:00—but not the *synchronous*-generation one, since hydro, geothermal, and cogeneration regulate voltage around the clock. One illustrative public case is the Cobb hydro station, divested by Transpower to Network Tasman around 2011: its pricing node was retired and its output now nets into the Stoke grid exit point (STK0661), a "lost node" whose metered reactive no longer represents a clean offtake. We flag these sites rather than discard them silently, and we show in Sections III-B and IV that the trajectory and the decomposition are robust to their removal: the contamination affects the absolute *level*, not the direction, intensity, or attribution of the drift.

F. Analysis methods

The trajectory is summarised by ordinary linear slopes for headline rates and, for robustness, by Theil–Sen slopes [24]: the median of the slopes through every pair of years (406 pairs for a 29-year series), an estimator that a few anomalous years cannot move, standard in long-record environmental trend monitoring, used here on both the national and per-GXP series; where the two estimators agree—as they do throughout—a trend rests on no small set of years. The test that the trend is not a metering artefact (Section III-B; workings in the extended version [23]) removes each GXP's own Theil–Sen trend and looks in what remains for a step shared across sites in the same year, using a change-point detector and a coherence statistic. The decomposition (Section IV) uses three methods: a clean-cohort comparison and a mixed-model cable dose-response with a company-cluster bootstrap (validated on synthetic data before use), which share a common underground-cable proxy and so are not independent of one another; and a first-principles cable-charging calculation, $Q = \omega C V^2 \ell$, with a $\pm 35\%$ physical band on cable capacitance carried through to the result, which fits no cable coefficient and is independent of the other two. All analysis is reproduced from public data in the accompanying notebooks.

III. The 29-Year Reactive Trajectory

A. Lagging to leading, overnight

Fig. 1 shows the central empirical result. Summed across the balanced demand panel, mean overnight reactive power fell from $+466$ MVAr (lagging) in 1997 to -451 MVAr (leading) in 2025, a near-monotonic linear drift of -31.1 MVAr/yr (robust Theil–Sen slope -31.3 MVAr/yr, 95% confidence interval (CI) $[-34.0, -28.1]$), crossing into net-leading overnight in 2016. On the wider 132-point balanced panel that also includes non-distribution connection points, the same drift is

$+673 \rightarrow -296$ MVAr at -30.3 MVAr/yr, crossing in 2020; the rate is essentially identical and the leading endpoint is, if anything, stronger on the demand-only panel once lagging industrial load is removed. The two panels are the same trend, labelled distinctly and never blended into one figure. Expressed per megawatt of overnight load, the drift is about -0.018 MVAr/MW/yr; we lead with this intensity and with the share of networks gone leading because, unlike the absolute MVAr total, they are insensitive to a few large contaminated nodes (Section III-B). The denominator is not doing the work: overnight load on the panel grew from about 1,680 MW in 1997 to about 1,960 MW in 2025. In the familiar comparator of Section II-A, the overnight aggregate power factor moved from 0.964 lagging to 0.975 leading—a numerically small excursion either side of unity concealing a 900-MVAr reversal, which is why the signed reactive power is the working variable throughout.

The diurnal contrast is itself a result. The evening peak crossed net-leading only in 2022 and remains far less leading than overnight, which locates the net-leading regime firmly at low load. Expressed as a fraction rather than a sum, the mean share of overnight half-hours that are leading rose from about 1% in 1997 to about 74% in 2025: overnight leading became routine rather than exceptional.

B. Measured, not an artefact; robust to contaminated sites

A reviewer's first question of any long series is whether the trend reflects the system changing or the measurement changing. Two features of the record answer that directly, and a further test closes the one gap they leave.

First, the drift does not rest on the older, less homogeneous early data. The metering and data pipeline modernised across the record (notably the move to the present Electricity Market Information system), and the one visibly suspect feature is an early step in the overnight leading fraction between 2000 and 2001, consistent with a change in metering vintage. Restricting to the post-2001 era of the present data pipeline removes any such concern, and the drift is if anything slightly steeper: -33.0 MVAr/yr over 2002–2025 against -31.1 MVAr/yr over the full record (the metering-fleet test proper—the 2014–2025 modern-fleet drift—is in Section II-C). A trend manufactured by early-era inhomogeneity would fade on the later data, not strengthen.

Second, a measurement artefact cannot reproduce the trend's physical structure. The net-leading crossing is specific to the overnight window—the evening peak stays lagging for years longer (Fig. 1)—while the deepening beneath the crossings is near-equal in the two windows over 2013–2025, with evening-peak load essentially flat (Section V-A): the signature of a standing capacitive term netted against different amounts of lagging load. The one systematic metering bias that does not average down scales with the metered load ($P \sin \beta$, bounded in Section II-C) and would deepen the heavily loaded evening window far more than the lightly loaded overnight one, which is not observed. Nor is the shift seasonal: recomputed separately for winter (June–August) and summer (December–February) months, the overnight aggregate drifts at -31.1 and -30.8 MVAr/yr respectively, so the trend is not an artefact of heating-season load composition. The drift also tracks each network's underground-cable intensity and is reproduced by a first-principles cable-charging calculation (Section IV); a metering error has no reason to correlate with cable kilometres or to emerge from the physics. Nor does a metering artefact at the grid exit point have a mechanism that scales with the customers behind it: the 2013–2025 deepening grows linearly with each grid exit point's connection count, with an intercept indistinguishable from zero (Section V-C). This combination of a diurnally structured, cable-correlated, connection-scaling, physics-reproducible shift is one that a change in measurement convention cannot counterfeit.

These two arguments address a gradual or staggered change, such as piecemeal meter replacement across sites. A single system-wide re-base is different: a one-date change of sign convention or central processing would step every supply point at once. We test for exactly that, detrending each site by its own robust line and asking whether sites step together in a single year, and find no such event: the change-point candidate years reach coherence of only 67% and 69% against a typical-year 62%, where a genuine re-base would drive coherence toward 100% (procedure and exhibit in the extended version [23]). The lagging-to-leading drift is therefore a measured property of the system, not an artefact of how it was metered.

A second objection, distinct from a metering artefact, is that the trend could be carried by a few grid exit points whose meters do not see a clean distribution offtake (Section II-E). We test it directly by removing them. Dropping the six *decoupled* grid exit points whose reactive power is set by plant or embedded generation (the *clean* cohort), and more strictly also the steady-generation and net-export points (the *clean-strict* cohort), we re-measure the trajectory on the balanced demand panel (the cohort exhibit is in the extended version [23]).

Two quantities behave very differently, and the difference is the point. The absolute total and the MVAr/yr slope *shrink* when the contaminated sites are removed—the 2025 total from -451 to about -300 MVAr and the slope from -31.1 to -23.4 MVAr/yr—because two of the removed nodes (Islington and Bromley) carry large, load-decoupled leading values, so the absolute *level* is partly inflated by contamination. The per-MW drift intensity, by contrast, is essentially unchanged (-0.0175, -0.0178, and -0.0180 MVAr/MW/yr across the full, clean, and clean-strict cohorts), and the share of networks net-leading overnight by 2025 is likewise stable (75%, 74%, 80%). The quantities that describe the phenomenon—its intensity per unit load and its near-universal reach—do not depend on the contaminated nodes; only the headline *size* does. This is why we lead with the per-MW drift and the share-leading migration and treat the absolute spine as a contamination-sensitive companion.

IV. Decomposition: Demand vs Network

The policy-relevant question is how much of the drift is the network's own cables, which inject leading reactive power as a distributed capacitance, and how much is the changing make-up of electricity demand itself: the retreat of inductive plant

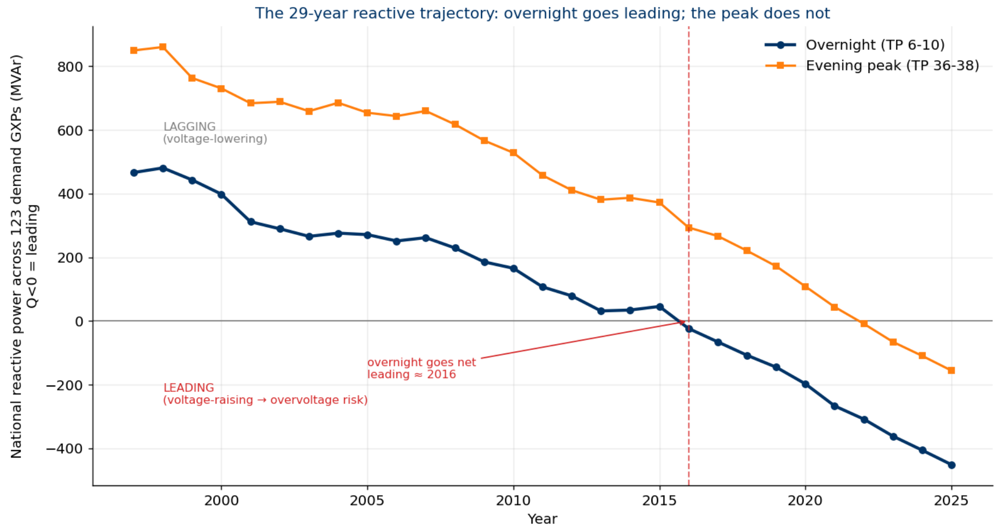

Fig. 1. The 29-year reactive trajectory on the balanced demand panel (123 grid exit points present in all years). Overnight (trading periods 6–10) drifts from strongly lagging ($Q > 0$, voltage-lowering) to clearly leading ($Q < 0$, voltage-raising), crossing net-leading in 2016; the evening peak (trading periods 36–38) follows much later. The diurnal split (the problem lives overnight at low load, not at the busy evening peak) is the signature of capacitive charging dominating when load is least able to mask it. Sign convention: $Q < 0$ is leading.

(simple motors) and the accumulation of power-electronic equipment (LED lighting, switch-mode supplies, heat pumps and other variable-speed drives). We call the second part *organic*: it is in the load fleet, not the wires. If the drift were mostly cables it would be a familiar network-planning problem; if it is mostly the demand fleet it is structural, nationwide, and outside any single network's control. We approach the question three ways; the two statistical methods share a common underground-cable proxy, while the physics-based method fits no cable coefficient and is independent of both. What the demand-side component physically *is*—the question the decomposition leaves open—is taken up in Section V.

A. Method 1: a clean-cohort contrast

History handed us networks at opposite ends of the cable spectrum. Some distribution businesses are almost entirely overhead line and have almost no underground cable; overhead line injects little leading reactive power, so these networks have almost no cable contribution. If their overnight reactive power still drifts leading, that drift can only be the demand fleet. We compare reactive power per MW of load (so network size cancels), load-weighted within each cohort (larger sites count proportionally more), for networks at or below 12% underground (*clean* in this method's sense: almost cable-free, distinct from the decontaminated cohorts of Section III-B) against networks at or above 40% underground (cable-rich); the method exhibits are collected in the extended version [23].

The clean cohort drifts from $+0.373$ to $+0.065$ MVAr/MW (slope -0.0101/yr), ending near balanced; the cable-rich cohort drifts from $+0.205$ to -0.306 MVAr/MW (slope -0.0134/yr), crossing well into leading. The ratio of slopes ($0.0101/0.0134 \approx 0.75$) puts the demand-side share of the cable-rich cohort's drift at about 75%, cable the remaining 25%. The reading rests on two assumptions the split depends on, stated plainly: that the organic per-MW drift is common across the cohorts (to the extent urban load fleets modernise faster than rural ones, the cable share is overstated), and that the at-most-12%-underground cohort carries no material cable term (to the extent it does, the demand share is overstated); the two possible violations pull the split in opposite directions. The exact ratio also moves with measurement choices, each worked in the extended version [23]: Schedule 9c's two circuit-length tables differ by a few points of %UG (percentage underground) and give about 75% ("total length for supply") against about 69% ("by operating voltage"), with Unison's cohort membership turning on the same choice; a series over all grid exit points rather than the balanced panel yields the wider, roughly 90/10 reading of earlier runs; attributing the network-changed sites (Section II-D) by their successor's cable dose throughout, the wires being physically continuous, moves the split to about 64% demand, the clean cohort being identical under every attribution; and removing the two decoupled Christchurch nodes (Section II-E) from the cable-rich cohort moves it to about 78% demand, 22% cable—the contamination, if anything, overstated the cable share. The *direction* is therefore robust while the *exact ratio* is not, which is why we treat Method 1's ratio as a hypothesis and bring two further methods. And although cable is the minority of the drift, it is the marginal factor that tips a network from balanced into net-leading: the clean cohort ends near balanced while the cable-rich crosses into leading, so cable intensity shapes *where* overvoltage appears first even though demand composition drives the trend.

B. Method 2: a validated dose-response

The second method gives every GXP a single number—the Theil–Sen slope of its overnight reactive power, scaled by its own typical size, through the 29 years—and regresses those per-site drifts on the site's cable dose (its network's percentage underground) in a mixed model with a random intercept per company (a company-level offset), with uncertainty from a cluster bootstrap (the fit re-estimated on many resampled datasets) that resamples whole companies, because sites under one company share a dose and are not independent. The fitted drift *at zero cable* (b_0) is the organic drift; the slope (b_1) is the extra drift per unit of cable. Before any use on the real data, the estimator was validated on synthetic panels with known planted answers, built on the real company structure and real cable doses and run through the identical machinery in two scenarios (a planted cable effect, and none); the full protocol and results are in the accompanying notebooks. The validation licenses a specific and limited claim: the estimator recovers the sign and magnitude of the organic drift reliably in both regimes, but it point-identifies (pins to a single number) the organic *share* only when the cable effect is itself identified—and on the real data it is not, so we rest no magnitude on this method. Line-fitting can neither manufacture a split here nor, with a cable effect indistinguishable from zero, pin one; the magnitude falls to Method 3.

On the real data, the organic drift is $b_0 = -0.016$/yr (95% CI $[-0.021, -0.010]$), leading in 100% of bootstrap resamples: unambiguous and dominant. The cable slope is $b_1 = -0.00005$/yr (95% CI $[-0.00037, +0.00012]$), which crosses zero—weakly identified, because cable dose is a company-level trait with only about 23 distinct values—so we deliberately publish no tight interval on the share (read across the cable range, the point estimate runs from 93% at the median dose to 78% at the most cabled networks, consistent with 100%; the point estimate is the single best-guess number before any range is put around it). Three robustness checks hold: adding capacitor intensity to the fit as a control (the cable effect read at fixed capacitor intensity) moves b_0 by only about 10% and leaves it strongly leading (consistent with the static-to-declining capacitor fleet of Section VI-B); the low correlation between cable dose and distributed-generation intensity (-0.22) indicates the organic signal is not an artefact of where embedded generation sits; and restoring the four sites dropped for want of a disclosing owner in the Section II-D mapping under their successor's cable dose moves b_0 by under 1%.

C. Method 3: first-principles charging physics

The third method computes the network contribution directly from physics, with no fitted coefficient, so it sidesteps the weak-identification problem entirely. Any energised length of cable or line behaves as a distributed capacitance and injects a fixed leading reactive power $Q = \omega C V^2 \ell$, where $\omega = 2\pi \cdot 50$ rad/s, C is capacitance per km, V the operating voltage, and ℓ the length. Underground cable has far more capacitance per km than overhead line. Using the published overhead and underground kilometres at each voltage from Schedule 9c, textbook per-km capacitances with an explicit $\pm 35\%$ band, and a band-by-band sum (charging scales with V^2, so high-voltage lines dominate), we compute the physical charging per network per year and subtract it from the measured overnight reactive power; the remainder is organic (Fig. 2). The band is set to span published per-km capacitance values across common distribution cable constructions and cross-sections, and it is carried through to the result rather than quoted and dropped. The organic remainder is a residual, and we name what else it absorbs: transformer magnetising reactive power (lagging and essentially static), series-reactance absorption (minimal at light load), and the distribution capacitor fleet (static-to-shrinking, Section VI-B)—terms that are second-order overnight or bias the residual toward lagging, so reading the residual as the demand-side drift is conservative.

Over 2013–2025 the measured overnight reactive power drifts at -49.9 MVAr/yr across all demand networks, of which the physical charging contributes only -9.2 MVAr/yr (cable length grows slowly) and the organic residual -40.7 MVAr/yr. The organic share of the *drift* is therefore about 82% (physical band 75–88%). Because this method fits no cable coefficient, it is independent of the two statistical methods; the division of labour is collected in Section IV-D. The window is set by the data: the disclosed circuit kilometres it runs on (Section II-D) are recorded consistently only from 2013, so the physics decomposes the 2013–2025 drift, about 45% of the record; extending the $\sim$80% share to the full 29 years is an inference, supported by the direction-consistent statistical methods (which do span the full record) and by the slow growth of cable length, not a second measurement. We stress that this all-network footprint rate (-49.9 MVAr/yr, 2013–2025) is a different dataset from the balanced panels of Section III (-31.1 MVAr/yr on the 123-GXP demand panel; -30.3 on the 132-GXP panel, 1997–2025) and from a separate cable-correlation analysis of the same public records (≈ 46 MVAr/yr); these measure different things over different samples and are not interchangeable.

A separate and more uncertain question is the composition of the leading reactive power *standing on the network today* (the level, as opposed to the drift). On the level the physics points the other way: a large, near-static cable and line charging baseline has always been present, and what changed is that the organic demand shifted from lagging—which used to mask the charging—toward leading, which no longer does. Today's net leading reactive power is then mostly long-standing charging *revealed* rather than newly created. The exact level split has a wide physical band and we do not quote a precise share for it; the level is a distinct, weaker claim and must not be confused with the drift result. We carry this only as the qualitative "uncovered charging" reading that informs Sections VI and VII.

D. Triangulation

Table I collects the three methods, which play different roles and are not all independent: Methods 1 and 2 both rest on the same underground-cable proxy, whereas Method 3 fits no cable coefficient at all. The reading follows that division of labour. The *direction* is robust and identified: the organic drift is leading in 100% of bootstrap resamples while the cable contribution to the drift is not statistically distinguishable from

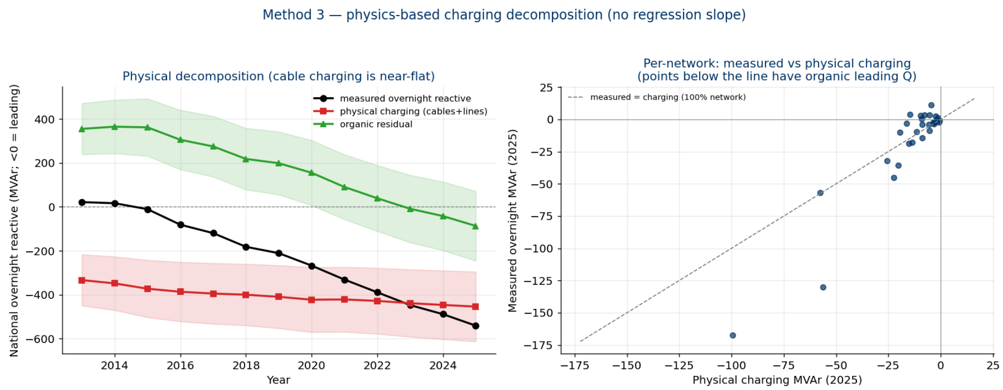

Fig. 2. Method 3, the physics-based decomposition over the all-network footprint (2013–2025, the years with consistent Commerce Commission circuit-length disclosures, from which the charging is computed). The physical cable and line charging (red, with the ±35% band shaded) is near-flat because cable length grows slowly, so almost all of the worsening trend in the measured overnight reactive power (black) is the organic residual (green). The drift is about 82% organic (physical band 75–88%). The right panel shows, per network, that most points lie below the unity line, i.e. carry organic leading reactive power beyond their charging.

TABLE I
THREE DECOMPOSITIONS OF THE DRIFT

Method	Establishes	Organic share of drift
1: clean-cohort contrast (cable proxy)	direction (shares the cable proxy with 2)	∼75% (78% with contaminated nodes removed; 64–90% across attribution, cable-intensity, and panel definitions)
2: dose-response (cable proxy)	direction / dominance	≥78%, consistent with 100% (cable term not pinned by the data)
3: charging physics (no fitted cable term)	**magnitude**	∼82% (physical band 75–88%)

zero, and together these imply the drift is dominantly demand-side. The *magnitude* is carried by the physics: the first-principles charging calculation, which fits no cable coefficient, puts the organic share at about 80% (physical band 75–88%). The two statistical methods are directionally consistent but, because the marginal cable effect (the extra drift per unit of cable) is not identified at the few company-level cable values available, do not themselves pin the share. We report the drift as dominantly organic and do not claim a point-identified split. Cable is the minority of the drift but the marginal factor in where net-leading territory is reached first. A cross-sectional check agrees: ranking the demand grid exit points by how leading they now are, the most-leading group is not the most-cabled—the sites with the steepest drift and the highest leading fraction sit at roughly half the cable intensity of the most-cabled group, and the gap widens further when the decoupled Christchurch nodes of Section II-E are removed—so cable cannot be what drives the deepest-leading behaviour, independently confirming the demand-side reading (the full cross-sectional typology is in the reproducibility notebooks).

V. The Demand-Side Term Identified: Standing Capacitance Behind Connections

The decomposition attributes the drift to the demand fleet; it does not say what in the fleet is doing it. Two measured properties narrow the answer, a specific physical candidate follows from them, and a scaling test probes the candidate's sharpest prediction.

A. Load-independent, around the clock

First, the drift is load-independent. Over the 2013–2025 window we summarise each grid exit point passing the full contamination screen (the clean-strict criteria of Section III-B, applied here to all sites metered in the window rather than to the 29-year balanced panel) by robust annual medians on wider windows than the Section II features (trading periods 1–12, midnight–06:00, and 35–39, 17:00–19:30). On the 114 such grid exit points with both endpoints metered, the median per-site change in reactive power is -2.25 MVAr overnight and -2.42 MVAr at the evening peak (median per-site ratio 1.06), while evening-peak load was essentially flat (median change $+3.4\%$). A change in how the connected load draws reactive power while operating would scale with the load operating—far larger at the peak than at 03:00—and so would the load-proportional metering bias bounded in Section II-C, whose spurious term goes as $P \sin \beta$. A near-equal deepening around the clock is instead the signature of *standing shunt capacitance*: always energised, indifferent to what the load is doing. It also reconciles the diurnal structure of Fig. 1: the two windows drift in parallel, and net-leading surfaces overnight first because the standing component is netted against more lagging load at the peak.

B. A physical candidate: the mandated EMC filter

Second, the capacitance must live behind essentially every connection, because the drift's reach is near-universal across grid exit points (Section III-B). We therefore advance a specific physical candidate, labelled plainly as a hypothesis: the electromagnetic-compatibility (EMC) filter that practically every mains-connected electronic product is required to carry. A switched-mode device chops current at tens to hundreds of kilohertz, and conducted-emission limits (the international CISPR limits, covering roughly 150 kHz–30 MHz, a condition of sale in most jurisdictions, New Zealand included) oblige its manufacturer to stop that noise at the appliance's own terminals; the universal compliance solution is a passive filter directly across the mains input (Fig. 3). Its elements carry safety-class names from where they connect: *X-capacitors* bridge line and neutral—the one position where a capacitor is permitted to be large (typically 0.1–2.2 μF), because a short circuit there is an overcurrent event the appliance's fuse clears rather than a shock hazard—while *Y-capacitors* run from the conductors to earth, where a short would electrify the chassis, so they must be built never to fail short and are held to nanofarads by touch-current limits; a common-mode choke completes the filter. To the 50-Hz fundamental this network is transparent except for the X-capacitance, which sits across 230 V drawing pure leading current whenever the plug is live: 0.1–1 μF injects 1.7–17 VAr, around the clock, because the filter sits ahead of the rectifier and of any standby logic—it must, since the noise it suppresses would otherwise leak even in standby. At 230 V the arithmetic is $Q = \omega CV^2$: 0.1, 0.47, 1, and 2.2 μF inject 1.7, 7.9, 16.6, and 36.6 VAr respectively. Commercial equipment carries larger input filters: a variable-speed drive holds tens of VAr and upward, energised wherever the installation is left connected. The regulatory structure makes the accumulation one-way: one instrument requires the capacitance in every product while, as Section I-A noted, no instrument bounds its fundamental-frequency injection—and a larger X-capacitor is generally the cheaper route to compliance. The hypothesis is disciplined by what it excludes: the large DC-link capacitor behind the rectifier is isolated from the fundamental once charged; motor-run capacitors sit in series with the motor's auxiliary winding and are disconnected with it; and rooftop solar inverters contribute almost none overnight (the isolation relay strands the converter-side filter capacitors, leaving only a small grid-side element). A bottom-up order-of-magnitude estimate—roughly two million households, each with 20–40 permanently connected filter-carrying devices at 2–4 VAr (roughly 80–320 MVAr in aggregate), plus a commercial and industrial fleet of the same order—puts the national standing total at roughly 150–600 MVAr, or 100–200 VAr per household as the central band. Sizing its most recognisable member bounds the composition: heat pumps were used in 66.8% of occupied private dwellings at the 2023 census, up from 47.3% in 2018—an adoption jump inside the study window, mostly at existing homes [25]—implying, at 1.2–1.6 units per equipped home, roughly 1.6–2.1 million residential units. At a datasheet-grade 0.3–2.2 μF of standing filter capacitance each (5–37 VAr), the class contributes roughly 8–78 MVAr: a material slice whose upper bound still sits at the floor of the residential envelope, so the bulk of the standing term belongs to the far more numerous small supplies. Because the filter, unlike the heating duty, is energised year-round, the slice is also consistent with the drift's winter–summer invariance (Section III-B); a heating-season mechanism would not be. The per-device values are physics and datasheet-grade figures, not New Zealand bench measurements, and we grade the attribution accordingly (component-level basis in the extended version [23]). The estimate's value is that it makes two testable

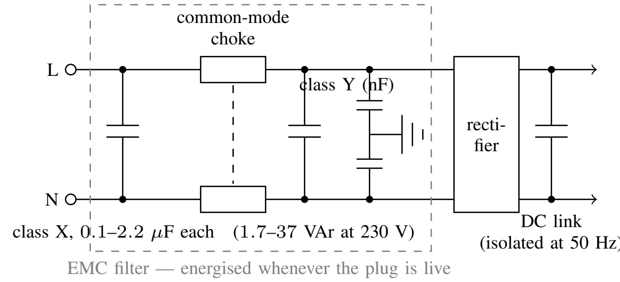

Fig. 3. The mains input stage of a switched-mode device. The EMC filter's X-capacitance sits directly across line and neutral, ahead of the rectifier and of any standby switching, and injects ωCV^2 of leading reactive power whenever the plug is live; the common-mode choke is transparent at 50 Hz, the Y-capacitors are nanofarad-scale, and the large DC-link capacitor is isolated from the fundamental behind the rectifier. Component values are datasheet-grade typical ranges.

predictions: the national scale of the drift, and—because the fleet lives behind consumer connections—that the drift should scale with the number of connections served, with nothing left at zero connections.

The first prediction is met at the order-of-magnitude grade the estimate carries. Summed across the same 114 grid exit points, the median overnight reactive power moved from $+188$ MVAr (net absorbing) in 2013 to -182 MVAr in 2025: a deepening of 369 MVAr (summarising each site by its 99th-percentile leading half-hour of the year, rather than its overnight median, gives 314 MVAr; the sites that deepened contribute 427 MVAr before the few that shallowed are netted off), equivalent to roughly 157–214 VAr per household across those variants. This is a bracket, not an allocation—the estimate bounds today's standing total, of which the twelve-year rise is the recently added part—but the measured rise sits inside the estimate's band, at the upper edge of its per-household centre. The second prediction is the more discriminating of the two; Section V-C tests it.

C. A connection-scaling test

Connection counts are public: the retail market-structure dataset on the same regulatory platform records the number of installation control points (ICPs—metered connections) behind each network supply point, monthly, from December 2003 [26]. We sum ICPs across retailers and embedded networks to each grid exit point, take calendar-year means for 2013 and 2025, and join them to the cohort under a documented correspondence audit, since the registry's root-NSP attribution and the metering's exit-point boundaries are not the same partition of the network: of the 114 sites, eight are direct-connect industrial points with no retail registry (a steel mill, a smelter, pulp and dairy plants), two lack a registry endpoint, one is unresolvable, and the remaining 103 all enter; sites whose registry codes demonstrably swap consumers with no matching movement of metered load (blocks of up to 27,000 move between neighbouring codes) are aggregated into station or sibling groups, giving $n = 96$ panel units. The test regresses each unit's 2013–2025 change in median overnight reactive power on the mean connection count it serves. The

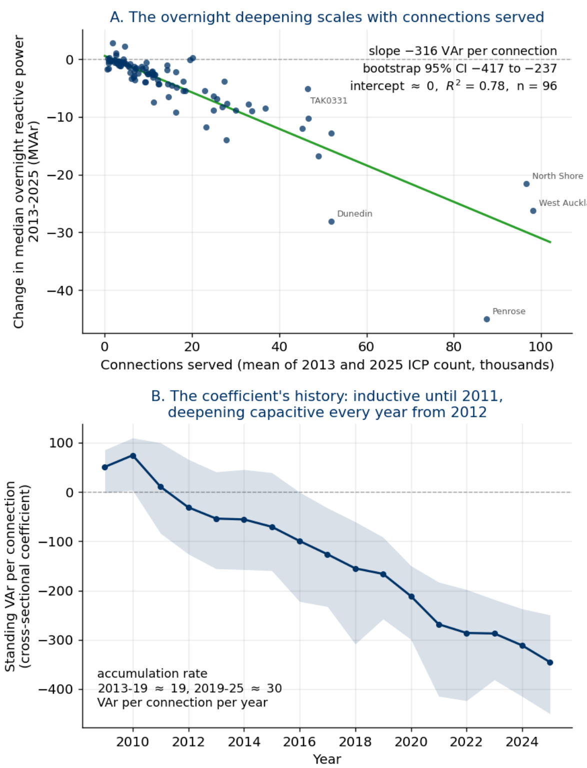

Fig. 4. The connection-scaling test. (A) Each screened panel unit's 2013–2025 change in median overnight reactive power against the mean number of connections (ICPs) it serves ($n = 96$): the change deepens at -316 VAr per connection (bootstrap 95% CI $[-417, -237]$) with an intercept indistinguishable from zero and $R^2 = 0.78$ (78% of the cross-unit variance explained); deleting any single unit moves the slope only within $[-341, -271]$. Nothing about a grid-exit-point metering artefact predicts an effect proportional to the connections behind the meter. (B) The coefficient's history: the per-connection coefficient refitted across units each year on the 92 units present throughout 2009–2025 (shaded: bootstrap 95% CI). On the inductive side of zero until 2011, it crosses in 2012 and deepens every year thereafter at an accelerating rate. Sign convention: negative = added leading (capacitive) reactive power.

decision rule was fixed before the first computation on 6 August 2026, is recorded in the accompanying notebooks, and is applied here unchanged: the hypothesis is supported if the bootstrap 95% confidence interval on the slope excludes zero, the slope magnitude lies within 50–400 VAr per connection, and the fit explains more than 15% of the cross-site variance; a grid-exit-point-level metering artefact, with no mechanism to scale with the connections behind the meter, predicts the mean-only model instead (the same average change at every site).

The result is Fig. 4. The deepening scales linearly with connections served: the slope is -316 VAr per connection (bootstrap 95% CI $[-417, -237]$), the intercept is -0.55 MVAr with an interval spanning zero—effectively nothing left at zero connections, and a through-origin fit agrees (-302)—and the fit explains 78% of the cross-unit variance. All three criteria are met, and the result is not fragile: a rank-based Theil–Sen slope agrees (-272), deleting any single unit moves the slope only within $[-341, -271]$ (removing the Penrose group, whose position pulls hardest on the fitted line, moves it to -271, a 14% shift with the verdict unchanged), and the mean-only artefact model loses by an AIC difference of about 143, a decisive margin on the standard model-selection score. The first run of the same pre-registered rule, on the narrower 83-site screen panel that predates the correspondence audit, reads -253 (CI $[-291, -192]$, $R^2 = 0.77$): the same verdict from a panel that under-samples the cable-heavy cities; we report both, never blended. A two-term variant separating existing connections from connections added over the window cannot split them on this panel: the two counts are strongly correlated across units and the added-connection term's interval spans zero, so the accumulation-versus-growth question falls to the coefficient's history below.

The coefficient also has a history (Fig. 4B). Refitting the across-units regression year by year on the 92 panel units present throughout 2009–2025—each year on the standing level rather than the 2013–2025 change, so the coefficient carries the whole connection-scaling class—the per-connection term starts on the *inductive* side of the axis ($+51$ VAr per connection in 2009, the motor-era fleet's residual signature), crosses zero in 2012, deepens every year thereafter (the interval is clear of zero on the capacitive side from 2017), and reaches -345 VAr per connection by 2025. The 2013–2025 movement, -291 VAr per connection, reproduces the two-endpoint slope within 10% by an independent route; the in-window accumulation rate is still rising (about 19 and 30 VAr per connection per year over 2013–2019 and 2019–2025, the 2009–2013 span being the crossing itself); and the series is smooth, where a staggered metering-replacement artefact would enter as steps (the largest single-year movements, over 2019–2021, coincide with the pandemic years). The emergence also overlaps the 2010–2013 meter-fleet replacement (Section II-C), and the coincidence is worth naming: an instrumentation change would enter as a step to a new level, not as a ramp that crosses zero mid-record and is still accelerating a decade after the rollout ended, equally in both diurnal windows, with nothing left at zero connections. The connection-scaling structure, in short, is not a fixture of the network: the fleet's per-connection signature swept from inductive through zero in the early 2010s and is still steepening on the capacitive side. We do not attribute the crossing date to any single device class.

Read honestly, the slope is the whole connection-scaling capacitance class, not the device fleet alone. Each connection also brings its share of medium-voltage cable (low-voltage cable is negligible, since charging scales with V^2), and the ICP counts carry no residential/commercial split, so commercial connections' larger filter fleets would ride in the same coefficient; that the slope sits somewhat above the 100–200 VAr central per-household band is therefore expected rather than anomalous. The units that sit off the fitted line do so in a sensible pattern: grid exit points serving industrial-weighted load (Takanini and Wiri, in south Auckland) deepen less than their connection counts predict, while the Penrose group, serving Auckland's cable-dense inner isthmus, deepens more. The scaling is also not site size in disguise. Load and connections rise together across sites, so a load-proportional metering error—the $P \sin \beta$ class of Section II-C—would mimic connection scaling; but that class is bounded far below the measured deepening, would concentrate the deepening in the loaded window rather than equally in both (Section V-A), and would over-predict precisely the industrial-weighted sites that in fact sit below the fitted line. What the test establishes at measurement grade is the scaling law itself; the split of the coefficient between device filters and per-connection cable remains hypothesis-grade. One rival survives in principle: a load-independent metering-registration drift phased in with meter replacement is heavily disfavoured by connection scaling with a zero intercept, but is formally excluded only by a measurement that bypasses the grid-exit metering chain. That measurement is identified and simple—night-time reactive flow at a residential distribution transformer with a known house count, where metering on the low-voltage side sees no transformer magnetising and no material cable charging, isolating the device fleet and splitting the coefficient—and is the stated next step.

VI. From Leading Reactive Power to Overvoltage

A. Why leading reactive power raises voltage

The relationship to hold in view is that the voltage change along a line depends mostly on the reactive power and the reactance, $\Delta V \approx (RP+XQ)/V$, with the XQ term dominant on the grid. When reactive power is lagging ($Q > 0$, the historical norm) voltage drops toward the load, and networks were designed around this with tap-changers and capacitors to prop voltage up. When reactive power is leading ($Q < 0$, the new overnight reality) the sign flips and voltage is pushed up. Overnight is the dangerous window because demand is lowest—so little load absorbs the reactive power or pulls voltage back—while the cable charging is always on. The mechanism can run away: charging rises with the square of voltage, so a higher voltage injects more leading reactive power and raises voltage further, and a protection trip that disconnects a transformer or load leaves the remaining network carrying more charging per unit of load, stressing the next element in turn. In isolated and autonomous systems this positive feedback is well documented [27], [28], and it is the cascade that the two 2025 blackouts below were found to be.

B. The system is already responding

The New Zealand system is visibly responding to overvoltage pressure from both ends.

Distribution capacitors are coming out. If networks were suffering low voltage they would be adding capacitors, which raise voltage; instead they are removing them. From the Commerce Commission disclosures, the national capacitor-bank count fell from 298 in 2015 to 254 in 2025, and the largest network removed 47 of 104 banks over the same period. The series is counts only and shows direction, not MVAr, but the direction is the point: a static-to-shrinking capacitor fleet cannot be causing a rising leading trend, and the removals are the signature of the opposite, high-voltage problem.

Transmission reactors are going in. Shunt reactors absorb leading reactive power, and the grid operator has been installing them in the Upper North Island: the Otahuhu reactor R472 (50 MVAr, 2022), the Pakuranga reactors R562 and R582 (100 MVAr each, 2023), the Wairau Road reactor R182 (63.7 MVAr, in service), and the Hamilton and Otahuhu STATCOMs (dynamic, 2023–2025), the ratings from Transpower's interconnection asset capacity disclosures and project records. Its System Security Forecast records both the practice and the remedy: "[o]vernight high voltages are currently managed by removing selected transmission circuits from service," and the two 100 Mvar Pakuranga reactors "will dramatically reduce the number of circuit switching operations required to manage high voltages under N-1 conditions" [29]. The operator's follow-up study confirms the outcome: with the Pakuranga reactors and the Hamilton STATCOM in service, "significant over-voltage issues have not been observed for current and future light load conditions" [30]. The investments, and the operator's own description of them, are consistent with managing an overnight overvoltage condition. (A further dynamic device, a Henderson STATCOM, is proposed under the WUNI Stage 2 major capex proposal but not yet approved.)

C. The realised risk, abroad

The regime is not hypothetical. The Iberian event of 28 April 2025 was found to be a voltage and reactive collapse: the Final Report identifies "non-effectiveness of voltage control" as a root cause, states that the excess of capacitive reactive power was the key point, and explicitly exonerates inertia [14]. The North Macedonia event of 18 May 2025 was a cascade of transformer trips from sustained overnight overvoltage, driven by the high capacitive reactive power of light-load transmission and distribution networks, with no renewables or compensation involved—so the cascade needs neither solar nor distributed generation [15]. Each event had local specifics; we read them as corroboration that the leading-reactive overnight regime is a realised blackout mechanism, with New Zealand's distinctive contribution the 29-year measurement and decomposition of how a system drifts into that regime in the first place.

D. Coverage of existing reactive-power provisions

New Zealand's reactive-power rules were written for the old problem of lagging load pulling voltage down at the daytime peak. The connection minimum power factor applies only during regional peak demand periods and sets a lagging floor with no leading limit and no overnight obligation [31], so the one window where overvoltage now lives is the window the rule does not cover; the reactive-demand measurement provision operates only on weekday hours, 07:00–21:00 [32]; and where lines companies charge for power factor, the published schedules charge on a lagging basis (e.g., [33]). We are careful about what this does and does not mean. The rules did not *cause* the leading surplus: Sections IV and V show the driver is the product-standard demand fleet, outside the rules' reach, and Section VI-B shows the capacitor fleet is shrinking, not the legacy of an over-correcting rule. What the existing provisions do is not cover, price, or limit the leading reactive power now accumulating overnight. Whether that coverage should change is a policy question outside this paper's descriptive scope; the descriptive finding is the measurement, and we note the policy question without arguing it.

VII. Implications for Power-System Analysis

The measured reversal relocates the binding voltage-security question. For the lagging era's failure mode—voltage collapse under heavy load—power-system practice holds a mature, layered stack: automatic under-voltage load shedding as a last resort, tap-changer blocking, PV- and QV-margin analysis, and decades of operational tooling, all built on load models of an inductive fleet. In the overnight regime measured here the equivalent layers are thinner: the planned compensation of Section VI-B exists and works, but the *automatic* last-resort layer acts in the wrong direction, and the margin tooling has no counterpart of comparable maturity. We take the asymmetry in three parts, add the load-model correction the measurement makes concrete, and close with its honest boundary.

The standard emergency lever points the wrong way. In a collapse, shedding load is corrective: every megawatt shed relieves the stress that is depressing voltage. In the light-load capacitive regime, load—even at near-unity power factor—is part of what holds voltage down: it absorbs real power and residual lagging reactive power, and its presence keeps the network's standing charge partly masked (Section IV-C). Shedding it removes that absorption and raises voltage. The Iberian Final Report lists exactly this as unavoidable physics—load shedding raised voltage [14]—and the Rhodes cascade showed under-frequency load shedding raising voltage and driving generators into deep reactive absorption, culminating in a loss-of-field trip that a small shunt reactor would have prevented [27]. The last-resort schemes discussed here were all designed for the other direction; none connects load or absorption when voltage runs away.

The driver resists dispatch, and the cascade reinforces itself. The distinction between the drift and the level matters here (Section IV): the *drift* into the regime is dominantly the demand fleet, but the standing overnight surplus the mechanism feeds on is substantially long-standing network charging that the fleet's shift has uncovered—distributed, always energised, scaling with the square of voltage (Section VI-A), and not itself dispatchable, since its removal means switching circuits out of service, spending N-1 security to buy voltage margin (Section VI-B). The cascade arithmetic is then inverted relative to collapse: each element lost in a voltage collapse—a stalled motor tripping, load disconnecting—reduces the demand that is depressing voltage, so the disturbance partly self-relieves, whereas each definite-time protection operation in an overvoltage cascade removes absorption and strengthens the driver, so the disturbance self-reinforces in quantised, irreversible steps. The Iberian sequence ran from first protection operation to blackout in well under a minute [14]; the North Macedonian event shows the same physics can equally grind through a lightly compensated system over hours [15]; the autonomous-system literature finds the positive feedback sharpest in small and weakly interconnected systems [27], [28].

The analysis stack under-represents the regime it is entering. Planning and protection studies still commonly represent demand with constant, lagging power-factor archetypes calibrated to the motor-dominated fleet, and expert reviews have already documented the error constant-power-factor load modelling introduces [6], [7]; the measured net overnight flow now carries the opposite sign, so the studies most responsible for revealing the risk are calibrated to a fleet that no longer describes the night. And while overvoltage protection and locally automatic shunt switching exist as device-level controls, the upper band lacks a coordinated counterpart to the distance-to-collapse toolkit: no routine margin quantity reports distance to capacitive runaway the way PV and QV margins report distance to collapse.

The measurement gives the load-model correction a concrete, portable form. The light-load fleet needs a standing shunt-capacitance term that scales with connections served—on the New Zealand panel, about 300 VAr per connection accumulated over 2013–2025 (Section V-C; a boundary-level figure that carries each connection's share of network cable with it)—alongside whatever power-factor archetype represents the operating draw. A constant-power-factor archetype cannot represent this term even qualitatively: it is present when the load is not; it accumulates behind existing connections rather than with demand growth, so it will not surface in load forecasts either; and it is young and moving—on the inductive side of zero as recently as 2011, its accumulation rate still rising (Section V-C)—so archetypes calibrated even a decade ago predate the term entirely.

None of this makes the regime unmanageable. With fast absorption installed, New Zealand's operator reports no significant overvoltage issues under current and future light-load conditions (Section VI-B): the gap is not feasibility but architecture—the automatic defences, the security-margin tooling, and the load models were built for the direction the system has left. Their redesign is development work beyond this paper's descriptive scope. What the measurement supplies is the direction and the rate: a national boundary flow that has crossed to the capacitive side of the axis around which those tools were built, driven dominantly by the load fleet, at about -0.018 MVAr per MW of overnight load per year, with the crossing already behind it.

VIII. Future Work

The measurement opens more questions than it settles; we collect the tractable ones.

The attribution measurement. The first priority is the decisive test named in Section V-C: night-time reactive flow at a residential distribution transformer with a known house count, which isolates the device fleet from the per-connection cable share, complemented by bench measurement of the standing injection of common device classes in standby. One instrumented night and one instrumented bench would move the central hypothesis of Section V-B from labelled to tested.

How long can it run? The coefficient's history (Fig. 4B) shows acceleration, not saturation. A single adoption curve must eventually saturate, but the fleet is a *succession* of curves—lighting, broadband equipment, heat pumps, and next an electric-vehicle charging and home-battery fleet expected to carry the largest input filters yet—so extrapolation calls for a renewal model of device generations—each wave layered on and later replaced—rather than a logistic (single saturating

S-curve) fit to the total. Bounding scenarios (the present $\sim$30 VAr per connection per year held constant; saturation at the current device census; the vehicle wave layered on top) built from appliance-stock and sales data would bracket the 2030s trajectory. Nothing in the record yet marks the turn.

Whether any instrument should see it. The standing term falls outside every instrument that could notice it—asset registers, load models, demand forecasts, product standards (Section I-A), reactive-power provisions written for lagging daytime load (Section VI-D); whether any should, and at which level (product standard, connection requirement, or network tariff), is standards development and regulatory economics beyond this paper's descriptive scope. We note only that the relative economics of absorbing the surplus at the grid level against not creating it at the product level appear unquantified in the open literature, and that the quantity now accumulates at a measured, forecastable per-connection rate.

Power-system analysis. Four concrete developments follow from Sections V and VII: a connection-indexed standing shunt-capacitance term and its placement within composite load-model structures; a routine margin quantity for the upper voltage band—distance to capacitive runaway—analogous to the PV and QV margins of the collapse toolkit; the frequency-domain counterpart, since distributed shunt capacitance reshapes the network's harmonic impedance as well as its fundamental-frequency balance; and the automatic last-resort defence architecture for the overvoltage direction, whose absence Section VII records.

Replication and the meter fleet as instrument. The test needs only settlement metering and connection counts, so any operator can reproduce it; comparative axes of interest include heating-electrified against gas-heated building stocks and systems whose device adoption leads New Zealand's, such as Norway's vehicle fleet. And where advanced meters record reactive energy at the connection itself, the per-connection term could in principle be read directly and continuously from the metering fleet, closing the chain from device to grid without new instrumentation.

IX. Conclusion

Using 29 years of regulatory half-hourly metering, we measured a national power system's reactive character shifting from lagging to leading at supply-point resolution: on the balanced demand panel (the same 123 grid exit points throughout), mean overnight reactive power fell from about $+466$ MVAr to about -451 MVAr (about -31 MVAr/yr), crossing net-leading overnight in the mid-2010s and at the evening peak only later—a measured phenomenon, not a metering artefact, holding in winter and summer alike, within the modern metering-fleet era alone, and with every contaminated grid exit point removed. The drift is dominantly the demand side: a first-principles charging calculation that fits no cable coefficient puts the organic share at about 80% (physical band 75–88%) over the 2013–2025 asset-disclosure window, with the two statistical methods corroborating the direction over the full record but not pinning the split. And the connection-scaling test identifies what the demand side is physically doing: at about 300 VAr per connection added over 2013–2025, with nothing at zero connections, from a term that sat on the inductive side of zero as recently as 2011 and is still accelerating, the load fleet is not merely drawing fewer lagging vars overnight; it is accumulating standing distributed capacitance of its own.

The significance of the reversal is what it does to the analysis. The defences, margins, and load models of power-system practice were built for lagging load and the under-voltage direction; the measured overnight flow has crossed to the other side, where the last-resort schemes act in the wrong direction and each protection operation reinforces the disturbance (Section VII); the same regime has produced two realised blackouts abroad, and the New Zealand system is already responding to it from both ends while its deeper demand-side driver remains unnamed in the rules. New Zealand's record—an advanced-state, small, isolated, heavily cabled analogue whose net-leading crossing preceded the 2025 European events by nearly a decade—is, to our knowledge, the longest direct measurement of that crossing, and both records the identification rests on are ones any operator already holds: if the standing term is carried by the globally standardised product fleet, every system connected to the same supply chain is accumulating it on a similar schedule.

The limitations are stated throughout: the balanced panel controls churn but does not eliminate it; the decomposition attributes the drift, not a causal mechanism in any single load; the exact organic share and the level composition are not point-identified; the device-level attribution of the connection-scaling capacitance remains a labelled hypothesis, with its decisive low-voltage measurement identified; and a single national system is one observation—offset, not cured, by its position as a leading-edge analogue and by the two independent European corroborations. The defence-scheme and load-model development this measurement motivates is separate work. What this paper offers is the measured trajectory, its decomposition, the term it identifies, and the direction power-system analysis must now build for.

Reproducibility

Every number and figure in this paper is computed from public data (Electricity Authority grid metering and Commerce Commission information disclosures) in an accompanying set of analysis notebooks, which also document the path from the raw public records. An extended version of this paper—including the complete measurement-error analysis, the system-wide re-base test, and the supporting method exhibits—accompanies the notebooks in the reproducibility repository [23]. Release of any underlying extract is subject to the data custodian's terms.

Acknowledgment

The author thanks colleagues at the Electricity Authority and Transpower for discussion of the reactive-power and voltage-management context, and the authors of the Rhodes autonomous-system studies for sharing corrected source material.

REFERENCES

[1] C. G. Kaloudas, L. F. Ochoa, B. Marshall, and S. Majithia, "Initial assessment of reactive power exchanges at UK grid supply points," in *Proc. CIRED Workshop 2014*, Rome, Italy, Jun. 2014, paper 0177.

[2] C. G. Kaloudas, L. F. Ochoa, B. Marshall, S. Majithia, and I. Fletcher, "Assessing the future trends of reactive power demand of distribution networks," *IEEE Transactions on Power Systems*, vol. 32, no. 6, pp. 4278–4288, Nov. 2017.

[3] RTE (Réseau de Transport d'Électricité), "Bilan de sûreté 2022," RTE, Paris-La Défense, France, Tech. Rep., Jul. 2023, sec. 5, pp. 26–27. Official English edition: *Reliability Report 2022*. [Online]. Available: https://assets.rte-france.com/prod/public/2023-07/2023-07-20-bilan-surete-2022.pdf

[4] Powerlink Queensland, "Managing voltages in South East Queensland," Powerlink Queensland, Project Assessment Conclusions Report, Sep. 2023. [Online]. Available: https://www.powerlink.com.au/sites/default/files/2023-09/Project%20Assessment%20Conclusions%20Report%20-%20Managing%20Voltages%20in%20South%20East%20Queensland.pdf

[5] J. Hannagan, R. Woszczeiko, T. Langstaff, W. Shen, and J. Rodwell, "The impact of household appliances and devices: Consider their reactive power and power factors," *Sustainability*, vol. 15, no. 1, p. 158, 2023.

[6] CIGRE Joint Working Group C4.24/CIRED, "Power quality and EMC issues with future electricity networks," CIGRE, Technical Brochure 719, 2018.

[7] Electric Power Research Institute, "Enhanced load modeling: Improved reactive power load modeling," EPRI, Palo Alto, CA, Technical Update 3002022354, Aug. 2021.

[8] International Electrotechnical Commission, "IEC 61000-3-2:2018+amd1:2020+amd2:2024 CSV electromagnetic compatibility (EMC) – part 3-2: Limits – limits for harmonic current emissions (equipment input current $\leq$16 A per phase)," IEC, Geneva, Switzerland, Tech. Rep., 2024, ed. 5.2 (Ed. 5.0 consolidated with AMD1:2020 and AMD2:2024).

[9] European Commission, "Commission regulation (EU) 2016/1388 of 17 August 2016 establishing a network code on demand connection," Official Journal of the European Union, L 223, 18 Aug. 2016, pp. 10–54, 2016, eLI: http://data.europa.eu/eli/reg/2016/1388/oj.

[10] M. Tazky, M. Regula, and A. Otcenasova, "Impact of changes in a distribution network nature on the capacitive reactive power flow into the transmission network in Slovakia," *Energies*, vol. 14, no. 17, p. 5321, 2021.

[11] A. H. Javed, P. H. Nguyen, J. Morren, and J. G. Slootweg, "Analyzing impact of capacitive reactive power flows at T-D interfaces," in *CIRED 2024 Vienna Workshop*. Vienna, Austria: Institution of Engineering and Technology, 2024, pp. 303–306.

[12] S. Baghban-Novin, M. Mahzouni-Sani, A. Hamidi, S. Golshannavaz, D. Nazarpour, and P. Siano, "Investigating the impacts of feeder reforming and distributed generation on reactive power demand of distribution networks," *Sustainable Energy, Grids and Networks*, vol. 22, p. 100350, 2020.

[13] A. Bosisio, F. Soldan, A. Morotti, S. Grillo, E. Bionda, and G. Iannarelli, "Lessons learned from Milan electric power distribution networks data analysis during COVID-19 pandemic," *Sustainable Energy, Grids and Networks*, vol. 31, p. 100755, 2022, also available as arXiv:2205.04894.

[14] ENTSO-E ICS Investigation Expert Panel, "Grid incident in Spain and Portugal on 28 April 2025 — final report," European Network of Transmission System Operators for Electricity, Brussels, Belgium, Tech. Rep., Mar. 2026, published 20 March 2026. [Online]. Available: https://www.entsoe.eu/publications/blackout/28-april-2025-iberian-blackout/

[15] ——, "Grid incident in North Macedonia on 18 May 2025 — final report," European Network of Transmission System Operators for Electricity, Brussels, Belgium, Tech. Rep., Apr. 2026, published 20 April 2026. [Online]. Available: https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Reports/2026/ENTSO-E_250518_Final_Report_260420.pdf

[16] D. Hume, "Power factors in the New Zealand power system," in *Electricity Engineers' Association (EEA) Conference*, Christchurch, New Zealand, Sep. 2025.

[17] Electricity Authority, "Electricity industry participation code 2010, part 10: Metering," Electricity Authority of New Zealand, Wellington, New Zealand, Tech. Rep., 2025, consolidation as at 1 April 2025; cl 10.37(2)(a) and Schedule 10.1, Tables 1, 5, and 6. [Online]. Available: https://www.ea.govt.nz/documents/7279/Part_10_-_Metering_-_1_April_2025.pdf

[18] International Electrotechnical Commission, "IEC 62053-22:2020 electricity metering equipment – particular requirements – part 22: Static meters for ac active energy (classes 0,1s, 0,2s and 0,5s)," IEC, Tech. Rep., 2020.

[19] ——, "IEC 61000-4-30 electromagnetic compatibility (emc) – part 4-30: Testing and measurement techniques – power quality measurement methods," IEC, Tech. Rep., 2015.

[20] Transpower New Zealand, "Asset management plan," Transpower New Zealand Limited, Tech. Rep., 2021, revenue-metering replacement carried out in RCP1 (2010–2013); legacy PSI Quad4 meters per the April 2010 AMP.

[21] International Electrotechnical Commission, "IEC 62053-23:2020 electricity metering equipment – particular requirements – part 23: Static meters for reactive energy (classes 2 and 3)," IEC, Tech. Rep., 2020.

[22] Institute of Electrical and Electronics Engineers, "Ieee std 1459-2010: Ieee standard definitions for the measurement of electric power quantities under sinusoidal, nonsinusoidal, balanced, or unbalanced conditions," IEEE, New York, NY, USA, Tech. Rep., 2010, revised as IEEE Std 1459-2025.

[23] D. Hume, "From lagging to leading: the measured emergence of standing capacitance behind consumer connections, 1997–2025 (extended version and reproducibility package)," GitHub repository with Zenodo snapshot, 2026, extended version carrying the complete measurement-error analysis, the system-wide re-base test, and supporting method exhibits, with the analysis notebooks that compute every number and figure from public data. [Repository URL and DOI to be inserted at submission.].

[24] P. K. Sen, "Estimates of the regression coefficient based on Kendall's tau," *Journal of the American Statistical Association*, vol. 63, no. 324, pp. 1379–1389, 1968.

[25] Environmental Health Intelligence New Zealand, "Home heating (indoor environment indicators)," Massey University indicator page, 2026, reporting Stats New Zealand Census data: heat pumps used in 47.3% (2018) and 66.8% (2023) of occupied private dwellings; accessed 7 August 2026. [Online]. Available: https://www.ehinz.ac.nz/indicators/indoor-environment/lack-of-home-heating/

[26] Electricity Authority, "Market share trends by root network supply point (NSP)," Electricity Market Information (EMI) market-structure dataset, 2026, monthly ICP counts by root NSP, December 2003–June 2026; retrieved 6 August 2026. [Online]. Available: https://www.emi.ea.govt.nz/Retail/Datasets/MarketStructure

[27] C. D. Vournas, V. C. Nikolaidis, and G. I. Tsourakis, "Coordinated countermeasures against overvoltage instability in autonomous power systems," *IEEE Transactions on Power Delivery*, vol. 36, no. 6, pp. 3329–3338, Dec. 2021.

[28] G. N. Psarros, G. I. Tsourakis, and S. A. Papathanassiou, "Dimensioning of reactive power compensation in an autonomous island system," *Applied Sciences*, vol. 12, no. 6, p. 2827, Mar. 2022.

[29] Transpower New Zealand (System Operator), "System security forecast 2022, part D: Power system security analysis — grid zone 2 (Auckland)," Transpower New Zealand Limited, Wellington, New Zealand, Tech. Rep., Dec. 2022, p. 12; the same passage appears in Grid Zone 1 (Northland), p. 11. [Online]. Available: https://static.transpower.co.nz/public/bulk-upload/documents/Part%20D%20GZ02%20Auckland.pdf

[30] ——, "N-1 thermal and voltage study — system security forecast 2024," Transpower New Zealand Limited, Wellington, New Zealand, Tech. Rep., Jul. 2024, version 1, July 2024 (minor update June 2026), p. 22. [Online]. Available: https://static.transpower.co.nz/public/bulk-upload/documents/N%20-%201%20Thermal%20and%20Voltage%20Study_Minor%20Update%202026%20June.pdf

[31] Electricity Authority, "Electricity industry participation code 2010, schedule 12.6: Default transmission agreement template," Electricity Authority of New Zealand, Wellington, New Zealand, Tech. Rep., 2026, connection Code, cl 4.4 (Minimum power factor: unity or 0.95 lagging by region, during regional peak demand periods only); consolidation as at 1 July 2026.

[32] ——, "Electricity industry participation code 2010, part 8: Common quality," Electricity Authority of New Zealand, Wellington, New Zealand, Tech. Rep., 2026, cl 8.67(4)(a): kvar measured weekdays 0700–2100 only; consolidation as at 1 July 2026.

[33] Powerco Limited, "Electricity pricing policy fy26," Powerco Limited, New Plymouth, New Zealand, Tech. Rep., 2025, §15: per-kVAr charge only below 0.95 lagging.